

Ian Salandy

Principal Software Architect | Distributed Systems, Performance & Reliability

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SUMMARY

Principal Software Architect with 15+ years of experience defining and scaling distributed systems architectures for high-throughput data platforms. Expert in capacity planning, cost-to-serve modeling, and system-level performance engineering across multi-billion event workloads. Proven ability to design architectures that scale predictably, reduce infrastructure cost, and improve reliability. Trusted advisor to engineering leadership for system risk, scalability strategy, and infrastructure investment decisions.

ARCHITECTURE & TECHNICAL LEADERSHIP

Distributed Systems Architecture & Design
Scalability Strategy & Capacity Planning
Cost-to-Serve & Infrastructure Efficiency Modeling
Performance Engineering & System Diagnostics
AI / Data Workload Performance Architecture
Benchmarking, Regression Governance & Standards
Performance Tooling & Platform Enablement
Cross-Team Architecture Leadership & Influence
Executive Communication & Risk Modeling

PROFESSIONAL EXPERIENCE

Salesforce — Acting Principal Architect / Staff Performance Engineer (2021 – Present)

- Defined scalability architecture for distributed data platforms processing billions of events annually.
- Designed system-level scaling strategies across compute, storage, and query layers improving throughput by 35%.
- Developed cost-to-serve models reducing infrastructure cost by 25% (~\$750K annually).
- Built predictive workload models forecasting system behavior under 2x–10x growth scenarios.
- Established performance diagnostics frameworks for cross-system bottleneck identification.
- Designed performance automation tooling enabling repeatable large-scale system validation.
- Led benchmarking and regression governance initiatives ensuring system stability across releases.
- Presented system risk, scaling constraints, and architectural recommendations to senior leadership.

IBM — Principal Performance & Capacity Engineer (2016 – 2021)

- Defined capacity planning and scalability strategies for enterprise distributed systems.
- Designed diagnostics frameworks to analyze system behavior under load.
- Developed infrastructure efficiency models to optimize cost and performance.
- Led JVM and backend optimization improving system latency and reliability.

Accenture — Lead Performance Engineer (2009 – 2016)

- Led performance architecture initiatives for large-scale enterprise systems.
- Designed load simulation frameworks to validate system scalability under peak conditions.
- Diagnosed and resolved distributed system bottlenecks impacting production reliability.

SELECTED ARCHITECTURAL IMPACT

- Defined architectures supporting billions of events across distributed systems
- Improved system throughput by 35% through architectural optimization
- Reduced infrastructure cost by \$750K+ annually via efficiency modeling
- Enabled predictive scaling and failure prevention through workload modeling

TECHNOLOGIES

Java, Python, SQL

Spark, Trino/Presto, Kafka

Kubernetes, AWS, EMR, EKS

Distributed systems design, workload modeling, performance optimization

EDUCATION

M.S. Computer Science — North Carolina State University

B.S. Computer Science — Morehouse College